

Machine Learning I: Hands-On 9

9.1 From ordinary datasets to meta-features

In ordinary supervised learning, each row usually represents one observation. In meta-learning, each row can represent an entire dataset or learning problem.

Consider the following five classification datasets. For each dataset, the table gives summary information about the original data.

Dataset	Class 0 examples	Class 1 examples	Raw features	Numeric features	Missing values	Total values
D1	10	10	2	2	0	40
D2	18	2	3	3	0	60
D3	6	4	10	10	5	100
D4	14	6	5	2	4	100
D5	8	8	4	3	8	64

Use the table to construct simple meta-features for each dataset.

1. For each dataset, compute the following meta-features:

- number of instances;
- number of raw features;
- numeric feature ratio;
- missing value rate.

Use

$$\text{Numeric Ratio} = \frac{\text{Number of numeric features}}{\text{Number of raw features}},$$

$$\text{Missing Rate} = \frac{\text{Number of missing values}}{\text{Total number of feature values}}.$$

2. For each dataset, compute the following class-distribution meta-features:

- majority class proportion;
- minority class proportion;
- imbalance ratio.

Use

$$\text{Majority proportion} = \frac{\max(n_0, n_1)}{n_0 + n_1},$$

$$\text{Minority proportion} = \frac{\min(n_0, n_1)}{n_0 + n_1},$$

$$\text{Imbalance ratio} = \frac{\text{Majority proportion}}{\text{Minority proportion}}.$$

3. Build a **meta-dataset** where each row corresponds to one of the five datasets and each column corresponds to one of the meta-features computed above.

4. Explain why this table is a meta-dataset. In your answer, clarify what the rows represent and how this differs from an ordinary supervised-learning dataset.
5. Based only on the meta-features, identify which two datasets you would expect to be more difficult for a simple classifier. Justify your answer using class imbalance, dimensionality, and missing values.

9.2 Meta-learning for algorithm selection

A full decision tree and a decision stump were evaluated on six classification datasets. A **decision stump** is a decision tree with maximum depth 1.

The table below presents three normalized meta-features for each dataset, along with the test accuracy obtained by both algorithms.

Dataset	Complexity	Imbalance	Missingness	Accuracy_DT	Accuracy_DS
D1	0.10	0.00	0.00	0.90	0.80
D2	0.15	0.90	0.00	0.80	0.90
D3	0.90	0.20	0.10	0.60	0.70
D4	0.10	0.00	0.00	0.95	0.65
D5	0.30	0.20	0.05	0.85	0.75
D6	0.25	0.00	0.15	0.75	0.75

Here, **Accuracy_DT** is the accuracy of the full decision tree, and **Accuracy_DS** is the accuracy of the decision stump.

1. For each dataset, compute the accuracy gap:

$$\text{Accuracy_Gap} = \text{Accuracy_DT} - \text{Accuracy_DS}.$$

2. Based on the accuracy gap, identify the best model for each dataset. Use the following rule:

- if the gap is positive, the best model is the full decision tree;
- if the gap is negative, the best model is the decision stump;
- if the gap is zero, the result is a tie.

3. Consider a new query dataset Q with the following meta-features:

$$Q = (\text{Complexity} = 0.12, \text{Imbalance} = 0.05, \text{Missingness} = 0.00).$$

Compute the Manhattan distance between Q and each of the six datasets D1-D6 using

$$d(Q, D_i) = |C_Q - C_i| + |I_Q - I_i| + |M_Q - M_i|,$$

where C is complexity, I is imbalance, and M is missingness.

4. Using **1-nearest-neighbor meta-learning**, recommend either the full decision tree or the decision stump for the query dataset Q .
5. Using **3-nearest-neighbor meta-learning**, recommend either the full decision tree or the decision stump for the query dataset Q by majority vote among the three closest datasets. In case of a tie, use the label *Decision stump or tie*.
6. Explain why this is a meta-learning problem rather than an ordinary classification problem. Your explanation should include:
 - what counts as an example;
 - what counts as a predictor;
 - what the target represents.

Conceptual focus

The unit of analysis has changed. The examples are now complete learning problems, the predictors describe those problems, and the target can represent which learning algorithm performed best. Meta-learning therefore learns *about learning tasks* rather than directly classifying their original observations.